

Assignment-2 Session13

Question 1: Create a Decision Table for a Flipkart-style coupon validation system where a coupon is only valid if:

- User is logged in
- Cart value is above ₹500
- Coupon has not expired

Answer

Conditions

C1: User Logged In

C2: Cart Value > ₹500

C3: Coupon Not Expired

Actions

A1: Accept Coupon

A2: Reject Coupon

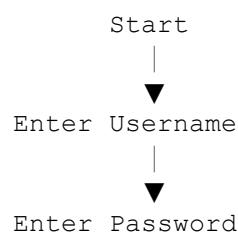
Decision Table

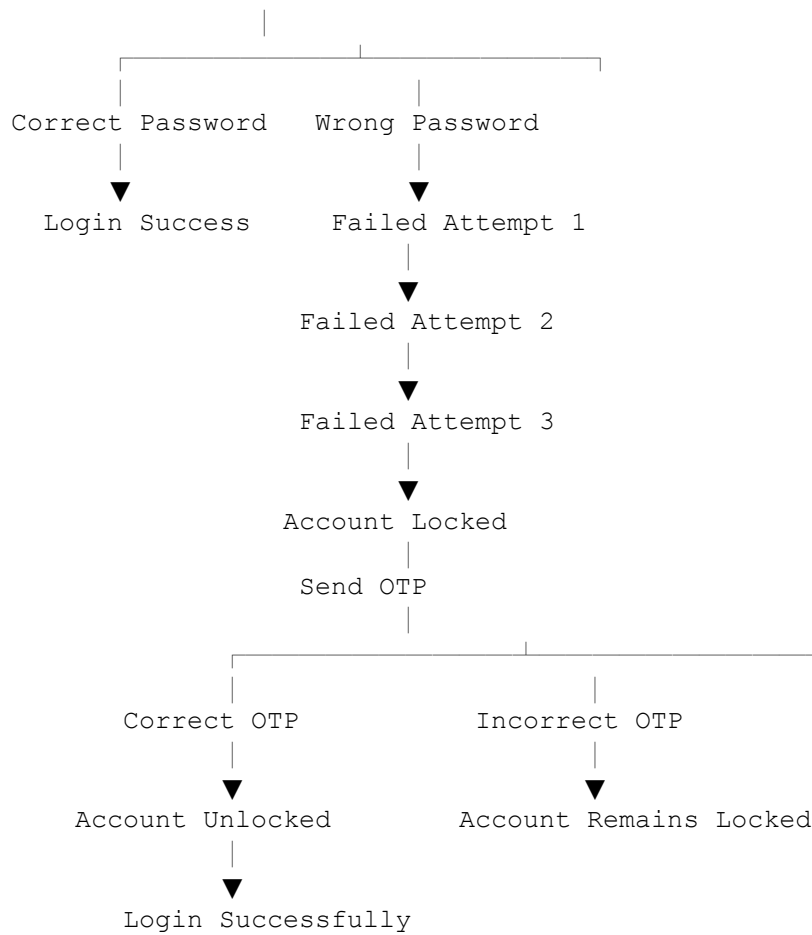
Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
User Logged In	Y	Y	Y	Y	N	N	N	N
Cart Value > ₹500	Y	Y	N	N	Y	Y	N	N
Coupon Not Expired	Y	N	Y	N	Y	N	Y	N
Action	Accept	Reject	Reject	Reject	Reject	Reject	Reject	Reject

Question 2: Draw a State Transition Diagram for a Paytm Login feature that locks the account after 3 failed password attempts and unlocks only after successful OTP verification.

Answer

State Transition Diagram





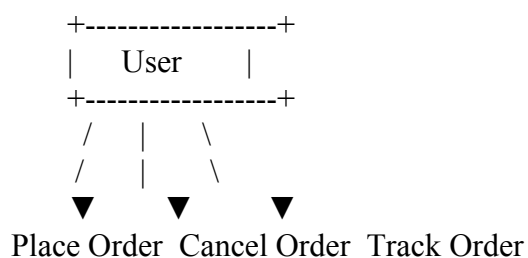
Test Scenarios

- ✧ Verify successful login with the correct password.
- ✧ Verify the account is locked after three consecutive incorrect password attempts.
- ✧ Verify a valid OTP unlocks the account.
- ✧ Verify an invalid OTP keeps the account locked.
- ✧ Verify the user can log in successfully after unlocking the account.

Question 3: Given the use case: "A Zomato user can place an order, cancel it before it is prepared, or track the order once prepared," create a Use Case Diagram and list at least 3 test scenarios.

Answer

Use Case Diagram



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(Before Food is Prepared)

Test Scenarios

Scenario ID	Test Scenario	Expected Result
TS-001	Place an order with valid payment details.	Order should be placed successfully.
TS-002	Cancel the order before restaurant preparation begins.	Order should be cancelled successfully.
TS-003	Try to cancel the order after preparation has started.	Cancellation should not be allowed.
TS-004	Track the order after preparation starts.	Live order status should be displayed correctly.

Question 4: Modify your Decision Table from Task 1 to include an additional rule:

The coupon cannot be used more than twice by the same user.

Answer

Additional Condition

C4: Coupon Usage Count ≤ 2

Updated Decision Table

Conditions	R1	R2	R3	R4	R5	R6	R7	R8	R9
User Logged In	Y	Y	Y	Y	N	N	N	N	Y
Cart Value > ₹500	Y	Y	N	N	Y	Y	N	N	Y
Coupon Not Expired	Y	N	Y	N	Y	N	Y	N	Y
Usage Count ≤ 2	Y	Y	Y	Y	Y	Y	Y	Y	N
Action	Accept	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject

Explanation

Previously, a coupon was accepted only when:

- ✧ User was logged in.
- ✧ Cart value exceeded ₹500.
- ✧ Coupon was not expired.